

LESSON 14 / 第十四課

Trees and Lightning: From Flash Density to the Conductor Path

樹木與雷電：從閃擊密度到導體路徑

Format. Short literature review. Claims that carry a number or a causal ranking are tied to a numbered source. Scale runs from the largest published pattern to the finest measured process. The lesson is not sequenced from the previous number and is not a regional case file unless the object itself is regional.

體例。短文獻綜述。帶數字或因果排序的陳述繫到編號文獻。尺度由最大已發表格局寫到最細已測過程。本課不由上一號遞推；除非對象本身是地域性的，否則不是單一地區檔案。

Abstract

摘要

Tree lightning protection is a preferred-path system. ANSI A300 Part 4 specifies copper conductors, air terminals and grounding beyond the drip line. Strike probability rises with height and isolation. An unprotected stroke follows wet cambium. Fasteners wound the stem; side-flash belongs in the target field. A designed path reduces the chance of a destructive cambial route. It does not prove zero injury or restored capacity.

樹木防雷是優選路徑系統。ANSI A300 第 4 部分指定銅導體、接閃器與滴水線外接地。擊中概率隨高度與孤立上升。無保護閃擊跟隨潮濕形成層。扣件傷害莖；側擊屬於目標場。設計路徑降低破壞性形成層路由的機會。它不證明零傷害或已恢復容量。

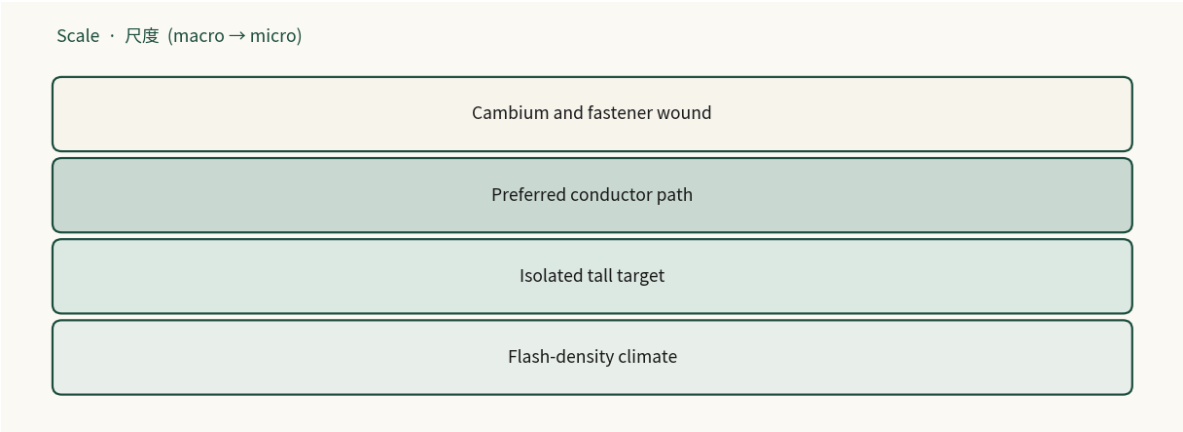


Figure 1. Nested scales from flash climate to the fastener wound.

圖 1. 由閃擊氣候到扣件傷口的嵌套尺度。

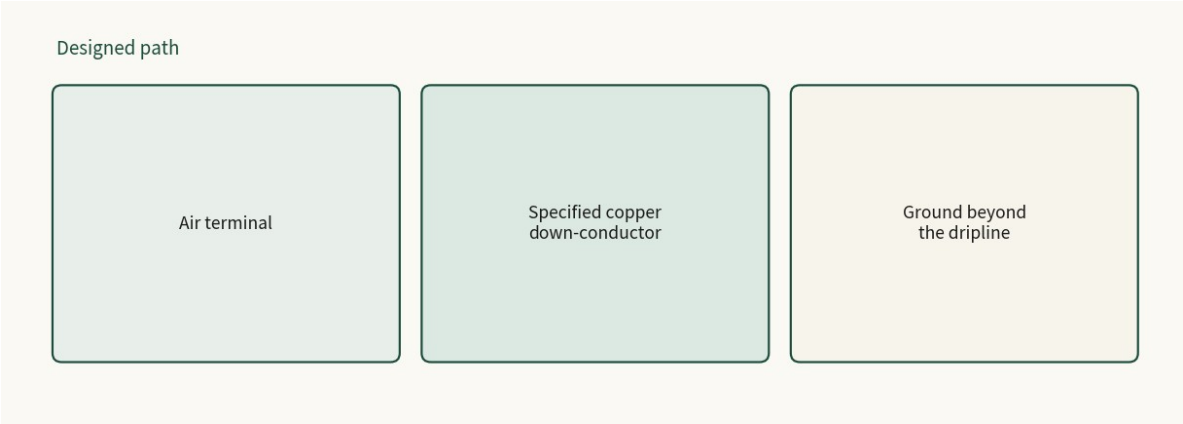


Figure 2. Air terminal, copper path, ground beyond dripline.

圖 2. 接閃器、銅路徑、滴水線外接地。

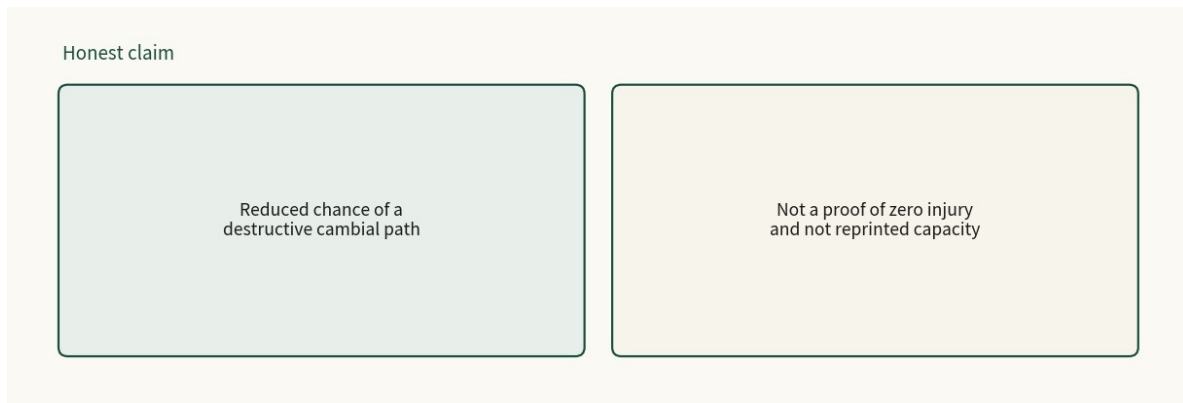


Figure 3. The honest claim is a reduced destructive path, not zero injury.

圖3. 誠實主張是降低破壞路徑，不是零傷害。

1. Scope

1. 範圍

The object is published tree lightning-protection specification and the injury physics it is written against.

對象是已發表的樹木防雷規格，以及它所針對的傷害物理。

2. Climate and path

2. 氣候與路徑

Strike probability rises with height and isolation [1]. Flash-density maps set how often the envelope is tested. An unprotected stroke follows the thin wet living cylinder [2]. ANSI A300 Part 4 specifies copper, air terminals, and grounding that leaves the drip line [2]. Side-flash to a person or a structure belongs in the target field of the same tree-risk sentence used everywhere else.

擊中概率隨高度與孤立上升 [1]。閃擊密度圖設定包絡被測試的頻率。無保護閃擊跟隨潮濕的薄活圓柱 [2]。ANSI A300 第 4 部分指定銅、接閃器，以及離開滴水線的接地 [2]。對人或結構的側擊，屬於處處使用的同一樹木風險句子中的目標場。

3. Wounds and claims

3. 傷口與主張

Fasteners are CODIT events. Conductors add a new conductive path and a new inspection object. The honest claim is a reduced chance of a destructive cambial path, not zero injury and not restored section.

扣件是 CODIT 事件。導體增加一條新導電路徑與一個新檢查對象。誠實主張是降低破壞性形成層路徑的機會，不是零傷害，也不是恢復截面。

4. Limits

4. 限度

Direct-strike experiments on large amenity trees are rare. Transfer of building-LPS rules onto living cylinders without the Part 4 constraints is not the published tree method.

大型園景樹上的直擊實驗稀少。沒有第 4 部分約束就把建築物防雷規則轉到活圓柱上，不是已發表的樹木方法。

Study questions

研習題

Choose the option that retains the published result. Answers follow the last item.

選出保留已發表結果的選項。答案在最後一題之後。

1. A tree LPS is:

1. 樹木防雷是：

A. Capacity restoration.

A. 容量恢復。

B. A preferred path.

B. 優選路徑。

C. 10-20-30.

C. 10-20-30。

D. P50.

D. P50。

2. Strike chance rises with:

2. 擊中機會隨何上升：

A. Flower colour.

A. 花色。

B. Height and isolation.

B. 高度與孤立。

C. Mulch brand.

C. 覆蓋物品牌。

D. i-Tree.

D. i-Tree。

3. Unprotected current follows:

3. 無保護電流跟隨：

A. Dry heartwood only.

A. 只有乾心材。

B. Wet cambium.

B. 潮濕形成層。

C. Soil nails.

C. 土釘。

D. QTRA.

D. QTRA。

4. Grounding should:

4. 接地應當：

A. Stop at bark.

A. 停在樹皮。

B. Extend beyond the drip line.

B. 延伸到滴水線外。

C. Use foil.

C. 用箔。

D. Equal t/R.

D. 等於 t/R。

5. A300 Part 4 metal:

5. A300 第 4 部分金屬：

A. Any scrap.

A. 任何廢料。

B. Specified copper.

B. 指定銅。

C. Lead.

C. 鉛。

D. Paint.

D. 漆。

6. Side-flash is part of:

6. 側擊屬於：

A. Soil pH.

A. 土壤 pH。

B. The target field.

B. 目標場。

C. P12.

C. P12。

D. Wall 2.

D. 第 2 壁。

7. Fasteners:

7. 扣件：

A. Cost-free.

A. 無成本。

B. Wound the stem.

B. 傷害莖。

C. Raise NSC.

C. 提高 NSC。

D. Set SOS.

D. 設定 SOS。

8. Honest claim is:

8. 誠實主張是：

A. Zero injury always.

A. 總是零傷害。

B. Reduced chance of a destructive path.

B. 降低破壞路徑機會。

C. No inspection.

C. 無須檢查。

D. Compact soil.

D. 壓實土壤。

9. Building-LPS rules copied onto trees without Part 4:

9. 沒有第 4 部分就把建築物防雷規則抄到樹上：

A. Are the published tree method.

A. 是已發表的樹木方法。

B. Are not the published tree method.

B. 不是已發表的樹木方法。

C. Raise P88.

C. 提高 P88。

D. Equal 3-30-300.

D. 等於 3-30-300。

10. Conductors add:

10. 導體增加：

A. Nothing to inspect.

A. 無可檢查之物。

B. A new inspection object.

B. 一個新檢查對象。

C. Instant Wall 4.

C. 立即第 4 壁。

D. A flare datum.

D. 幹基基準。

Answers / 答案 1-B 2-B 3-B 4-B 5-B 6-B 7-B 8-B 9-B 10-B

References

參考文獻

[1] Taylor A.R. and lightning-tree reviews; flash-density climatology.

[2] ANSI A300 Part 4; ISA BMP lightning protection.

[3] Smiley E.T., Lilly S. tree lightning-protection practice texts.

Closing note. This lesson is written as a short review article. Disagreement in the literature is kept in view where the results actually diverge.

結語。本課以短綜述文章寫成。文獻真正分歧之處，兩造結果都保留在視野內。